Backend Engineer, Addis Ababa, Ethiopia

DAWIT ELIAS

X Twitter/ X

LinkedIn

dawiteliaskassaye@gmail.com

www.dawit-elias.vercel.app

My Github

WHO AM I ?

HI !

As a highly skilled Software Engineer with over 2 years of experience, I have developed a fairly profound expertise in software development, system architecture, and project management. My career has been marked by a commitment to leveraging innovative technologies to create robust and scalable solutions that drive business success. With a strong foundation in both backend and frontend development **(mostly backend)**, I possess a comprehensive understanding of the software development lifecycle. I am adept at leading cross-functional teams, mentoring junior developers, and collaborating with stakeholders to deliver high-quality products on time and within budget.

EDUCATION AND CERTIFICATIONS

BSc in Computer Science — **Feb 2021 - present**
University Of Gondar

Harvard CS50 course series — **July23 , 2022**

ALX Ethiopia batch of 2023 (drop out because of family issues) — **January 09, 2023**

Internships

Camara Education - backend intern — **Feb 2023**

Prodigy InfoTech - Fullstack web dev intern

TECHNOLOGIES I USE

- **Frameworks (Frontend) / Design:** TypeScript, React, Next.js, Redux, Ant Design, Angular, Svelte Kit, Material UI, SCSS/SASS
- **Frameworks (Backend)**: Gin, Echo, Gorilla mux, Next JS, Node.js, Express.js, PostgreSQL, MongoDB, Nest.js

- **DevOps/Tools:** Git, AWS (EC2, S3, Lambda, CloudFormation, CloudFront), Azure (Virtual Machines, Blob Storage, Functions, App Service), Playwright, Zapier, Docker, Kubernetes, Jenkins, CircleCI, Travis CI, Ansible, Terraform, Puppeteer
- **Testing:** Golang testing package, testify, Mocha, Chai, Jest, Cypress, Selenium, Protractor, JUnit, PyTest
- **Programming Languages:** Golang, Rust, TypeScript.

- **Software Development:** Domain-Driven Design (DDD), Test-Driven Development (TDD), Behavior-Driven Development (BDD), Microservices, Service-Oriented Architecture (SOA), Event-Driven Architecture (EDA), Unified Modeling Language (UML), Agile (Scrum, Kanban), DevOps (CI/CD, Infrastructure as Code), Serverless Computing.

- **Operating Systems**: Windows, Linux, macOS Platforms: AWS Cloud, Azure Cloud, Google Cloud Platform (GCP), Digital Ocean

- **Tools:** Visual Studio, Vs Code, IntelliJ IDEA, Eclipse, Postman, GitLab, Bitbucket, Jira, Confluence, Slack, Trello, MS Office 365, Sharepoint API: RESTful API, GraphQL API, gRPC, WebSocket
- **Knowledge:** Real-time systems, WebRTC, Parallel computing, Networking, Functional programming, Object-oriented programming, Design Patterns, Apache Pulsar, NoSQL, Elastic Search, Kibana, Logstash, Beats, Grafana, Prometheus, Kafka, RabbitMQ, Redis, Memcached, Kubernetes Operators, Istio, Helm, Serverless Framework.

- **Databases:** MongoDB, PostgreSQL, MySQL, Redis, DynamoDB, MS SQL Server, Oracle Database, Couchbase, Cassandra

GO GOLANG SPECIFIC WORK EXPERIENCE

FREELANCING & PERSONAL PROJECTS **Feb 2022 - present**

- **Implemented a Distributed and secure File Server (personal project):** I designed and developed a highly scalable, secure, and fault-tolerant file storage system, improving data accessibility, security , and reliability across multiple nodes for a government organization using Golang from scratch without using external libraries for the sake of avoiding security risks related to using them but the project got declined.

- **Realtime chat app backend for a school *(Freelance project)*:** developed a real-time chat application for a school based Addis Ababa using Golang, leveraging WebSockets for real-time communication. The application allows students to join chat rooms with their colleagues as well as tutors, send messages, and see updates in real-time.

- **Collabor8 - a Real-time collaboration whiteboard (personal project)** : initiated by the interview I landed for golang backend position at career.io, started working on this project using Gin, gRPC, WebRTC, PostgreSQL, Railway, Docker for the server side and Next JS for the client. I will release this for production use on Jan, 2025.

- **API Gateway with Rate Limiting and Authentication*(personal project):*** using Golang, Redis (for rate limiting and caching), PostgreSQL (for storing user credentials and configuration), OAuth2 for authentication, I built an API gateway that handles authentication, rate limiting, and request forwarding to different microservices. The gateway is configurable to add new microservices and set rate limits.
- **Backend payment and order service *(take home project)* -** Developed 5 services, **Notification**, **payment** , **order** ,**wallet** , and **logger** service along side a **gRPC** server and **API** gateway and a **Next JS** front end checkout page for a project to land a full-time golang backend engineer position at **Addispay Financial Technologies SC.**.

PROFESSIONAL WORK **Jan 2024- present**

ADDIPAY FINANCIAL TECHNOLOGY

- **Resolved Service Discovery and Deployment Issues :** Identified and addressed bottlenecks in service discovery mechanisms, migrated the system from HashiCorp consul to Kubernetes, and streamlined the deployment pipeline using Docker and Kubernetes, reducing downtime and enhancing reliability.

- **Developed a transactions API, account service, and Merchant Gateway :** Leveraging the concurrency patterns and parallel processing Golang offers, I created a secure and efficient API that handled numerous transactions daily, enhancing transaction speed and security. I used Apache pulsar as the broker and redis for caching.

- **collaborated in the integration of kacha payment processor to the payment gateway**
- **Migrated a large MySQL database successfully to production environment**
- **Implemented data aggregation APIs for analytics using mongoDB and pure golang**

- **Deployed and Managed a Notification Service for the Payment Processor:** Implemented a real-time notification service using Redis for caching, and pulsar for message brokering, significantly improving user engagement and communication efficiency.

BANKERR - MY STARTUP

- **Bankerr**: In my country Ethiopia how people create bank accounts is either going to the bank or when a random bank clerk approaches them to create the account for them and get a commission. Eventually, people got tired of getting approached by these clerks and having to go through these processes to the point where they started ignoring people approaching them for other things. So me and colleagues of mine, **Temam Awleu** and **Ezedin Fedlu**, the former, a senior software engineer working at **Awash Bank**, one of the largest banks in Ethiopia, and the later, a senior software engineer at **Volvo cars,** have started working on a platform on which people can create and manage all of their bank accounts sparing them from this headache. We are currently building the MVP using Golang + gin for the backend and Next JS + typescript for the frontend with mock APIs on swagger and waiting the approval for access to the necessary APIs for the build.

- You can see the demo here => bankerrsolutions.com

P.S. The codebases for the projects listed above and other projects is on my github page

My Github

LinkedIn X Twitter/ X dawiteliaskassaye@gmail.com www.dawit-elias.vercel.app